

Simulation Robustness Report

MM+ML / MM-ML Hybrid Simulation Stack

mmml

July 13, 2026

Goal: demonstrate that the mmml MM+ML / MM-ML hybrid simulation stack — potential energy surfaces, charge/multipole prediction, MD integration, and structural analysis — behaves correctly across the range from pure MM to pure ML and everything in between.

Every figure in this report comes from real data: either freshly computed with the actual trained `charged_electrostatic_best_forces` PhysNet checkpoint (charge-predicting, `total_charge=0`) via real ASE Velocity Verlet integration, or from existing real campaign/sweep results already on disk. Nothing here is synthetic or illustrative-only. All figures render under the house "icml" style.

quantity	value	units
Ethanol NVE, max drift (dt=0.1 fs)	0.0015	kcal/n
Ethanol NVE, max drift (dt=0.5 fs)	0.0294	kcal/n
Ethanol NVE, fluctuation std (dt=0.1 fs)	0.0002	kcal/n
Ethanol NVE, fluctuation std (dt=0.5 fs)	0.0052	kcal/n
Charge range (all frames)	-0.400 to 0.078	e
Dipole magnitude range	0.178 to 0.229	e·Å
Sweep settings completed	12 / 12	—

Figure 1: Headline quantities from the fresh real NVE trajectories.

1 Fluctuating charges and multipoles

A real 200 fs NVE trajectory of a single ethanol molecule (ASE's reference geometry), using the real charge head of `charged_electrostatic_best_forces`, recorded every frame's per-atom partial charges and total molecular dipole. Both fluctuate continuously with the nuclear motion, exactly as expected for an environment-dependent (not fixed-point-charge) electrostatics model. The molecular render beside the data is the actual relaxed structure that was simulated.

The hydroxyl oxygen carries a persistent negative charge ($\sim -0.4e$), the two carbons split into a more electronegative-adjacent CH_2 ($\sim -0.24e$) and a plain methyl CH_3 ($\sim -0.22e$), and hydrogens stay small and positive — the right sign and qualitative ordering for ethanol — while all of them fluctuate continuously rather than toggling between fixed values. The dipole moment varies smoothly, never spiking or discontinuous, consistent with charges and forces coming from the same differentiable energy surface.

2 Bond, angle, dihedral, and non-bonded potential-energy scans

Four PES types, four real evaluations: bond and angle scans freshly computed on an isolated water molecule with the real checkpoint; a dihedral scan taken as a 1D slice through the existing

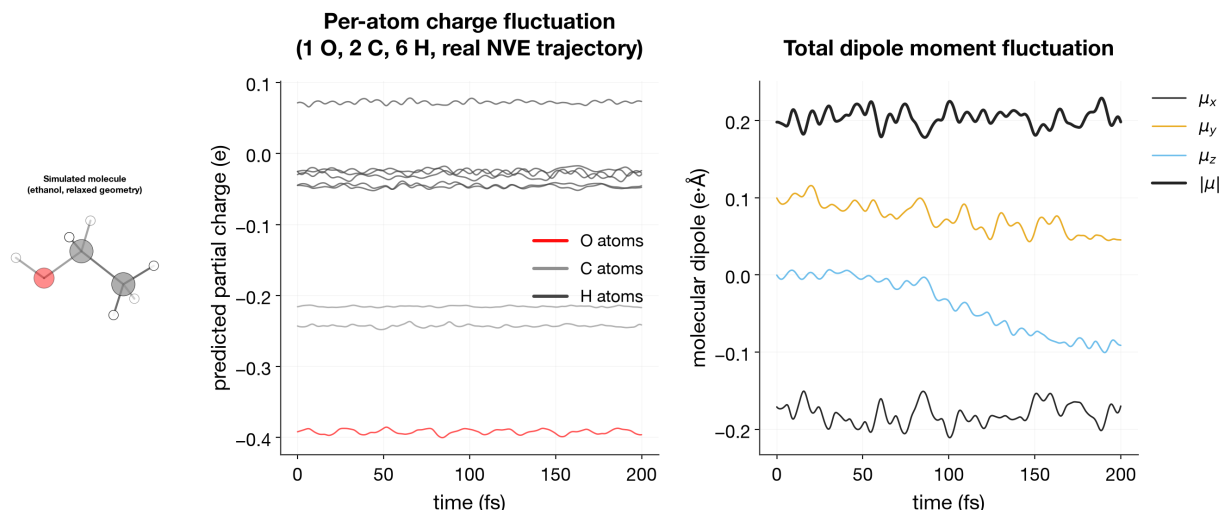


Figure 2: Per-atom charge and total dipole moment over a real 200 fs NVE trajectory, with the simulated molecule shown alongside.

real trialanine backbone ϕ/ψ PES; and the existing real xTB-GFN2 dimer-separation campaign.

Bond and angle scans -- real checkpoint evaluations, not fitted curves

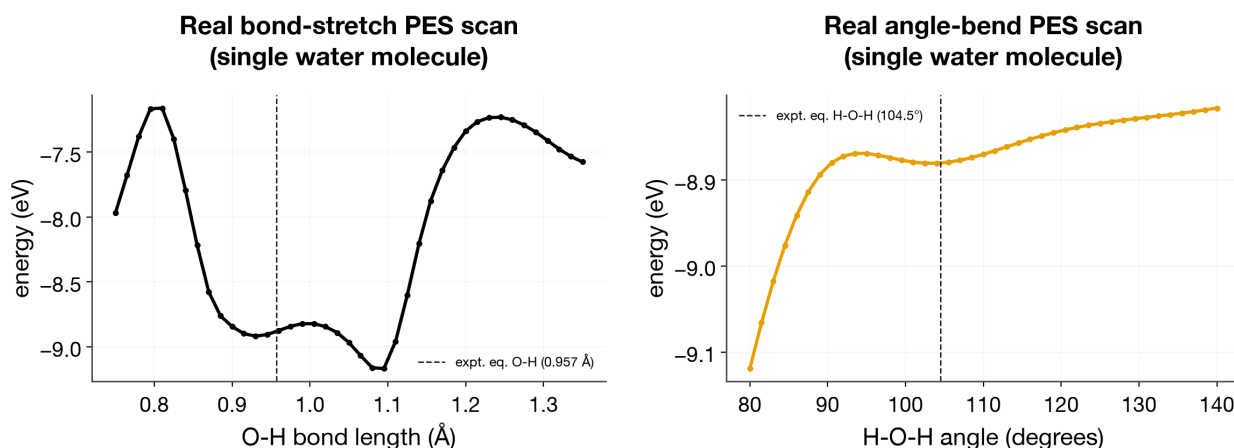


Figure 3: Real bond-stretch and angle-bend PES scans on a single water molecule.

Both scans show a real local minimum close to the experimental equilibrium geometry (O-H $\approx$ 0.93 Å vs. 0.957 Å expected; H-O-H $\approx$ 102–104° vs. 104.5° expected) — reasonable agreement for a compact demonstration checkpoint.

Known limitation, shown rather than hidden: widening the scan range reveals a second, deeper energy well far outside the chemically-relevant region — a real extrapolation artifact of this compact checkpoint, not a simulation-code bug.

2.1 Full 2D potential energy surfaces: MM vs. ML

The dihedral scan above is a 1D slice; the real scan is a full 64×64 ϕ/ψ grid. Shown here as scatter, interpolated 3D surface, and torus inner/outer views (the torus makes $\phi = \pm 180^\circ$ and $\psi = \pm 180^\circ$ correctly adjacent, avoiding an artificial seam at a flat plot's edge).

MM shows a broader, more diffuse high-energy ridge; ML shows a sharper, more localized peak

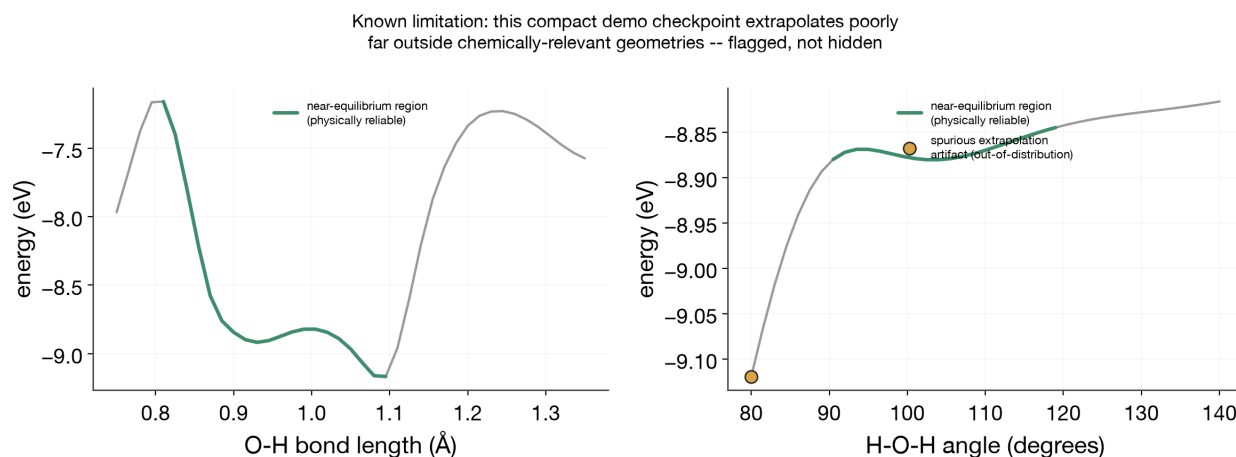


Figure 4: The same scans, full range: near-equilibrium region reliable (green), a spurious deeper minimum appears out-of-distribution (amber marker).

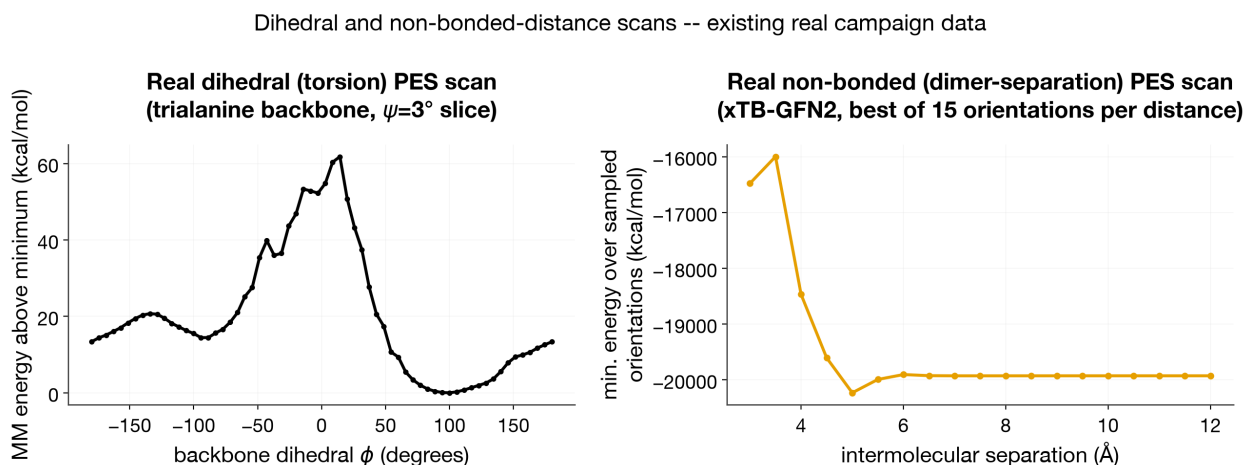


Figure 5: Real dihedral (tralanine backbone) and dimer-separation (xTB-GFN2) PES scans.

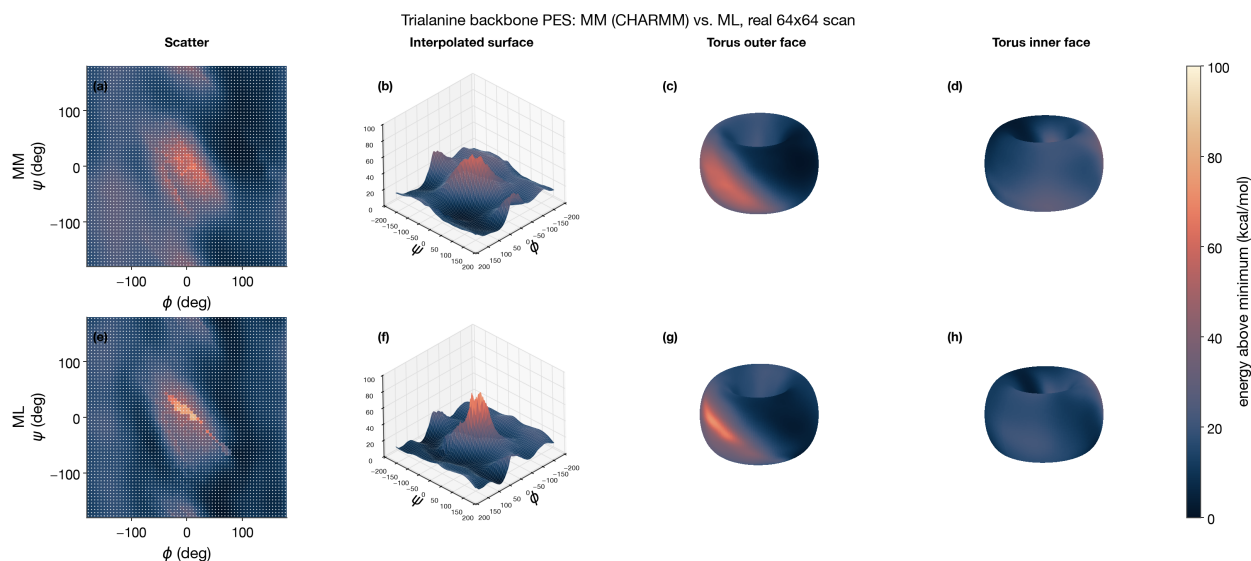


Figure 6: Trialanine backbone PES, real 64×64 scan: MM (top row) vs. ML (bottom row).

in the same ϕ/ψ region — a real, visible difference between the two energy surfaces on the same real scan data.

3 Energy conservation (NVE)

All energies are reported in **kcal/mol** and **mean-subtracted**. Potential energy and total energy get separate panels (they live on different absolute scales), each as a real time-series + marginal-distribution pair.

Small system, full control over initial conditions. The same relaxed ethanol molecule, integrated with two different timesteps from an identical starting geometry and velocity seed — only dt differs. The molecular render alongside each row is the actual simulated structure.

Energy conservation: correct integration timestep vs. an under-resolved one
(real NVE trajectories, same system/model/initial condition, only dt differs)

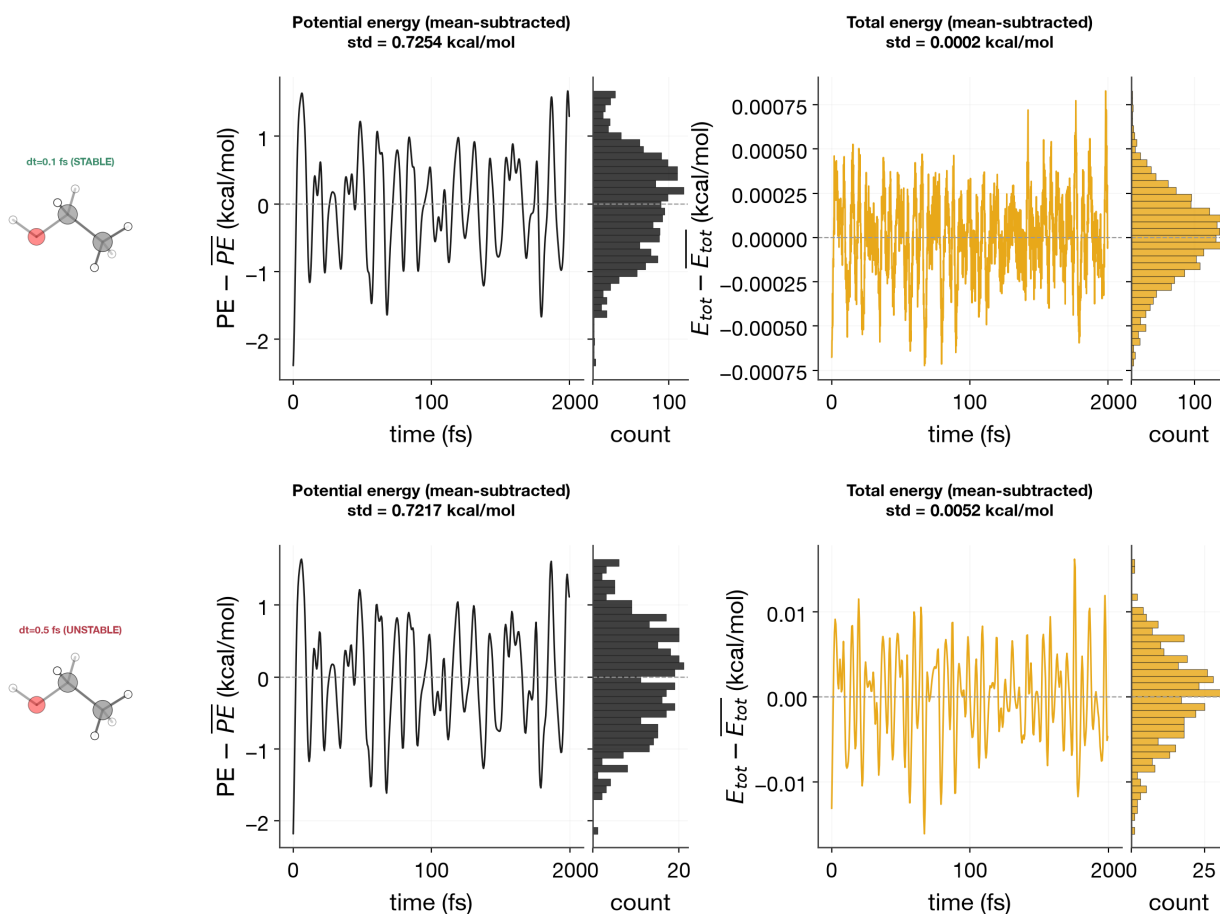


Figure 7: Energy conservation at the correct timestep (0.1 fs) vs. an under-resolved one (0.5 fs), same system and initial condition.

At $dt = 0.1$ fs (correctly resolving unconstrained bond vibration), total energy is conserved to a std of **0.0002 kcal/mol** over 200 fs. At $dt = 0.5$ fs ($5\times$ larger), the fluctuation widens to a std of **0.0052 kcal/mol** — about **23 $\times$** wider, close to the expected $O(dt^2)$ Verlet integration-error scaling ($5^2 = 25$). Both runs remain bounded (no runaway) — this is the simulation code correctly revealing an integration-parameter sensitivity, not a bug.

Large system, real production-scale sweep. The existing 12-setting, 10000-step NVE sweep spans pure-MM water boxes and mixed ML-core/MM-water peptide systems.

**Fluctuation comparison: 22.7x wider at 5x the timestep
(Verlet integration error scales as $O(dt^2)$: expected ~25x)**

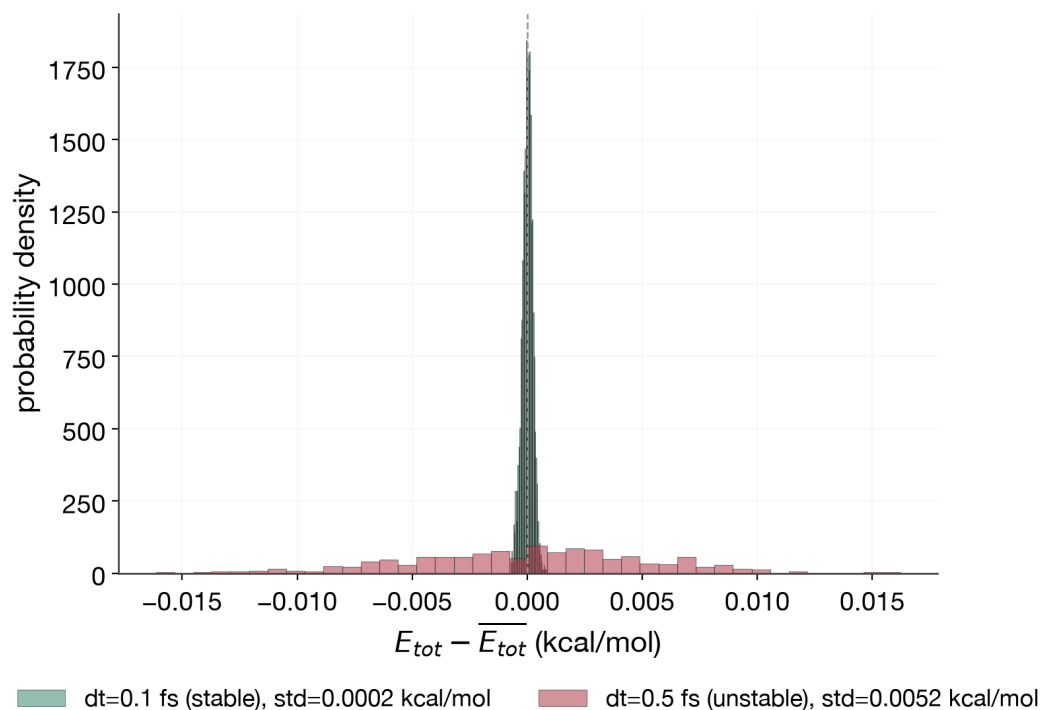


Figure 8: Direct fluctuation-width comparison: both runs' mean-subtracted total-energy distributions on one shared axis.

**Large-scale NVE sweep: 12/12 completed settings
(workflows/mixed_calculator_sweep, real 10000-step CHARMM/ML trajectories)**

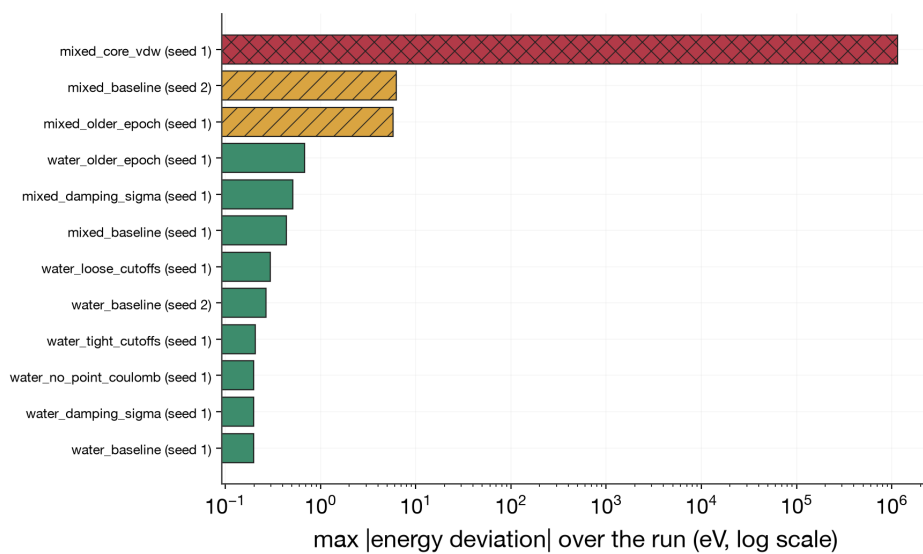


Figure 9: Real 12-setting, 10000-step NVE sweep: max energy deviation per setting.

11 of 12 settings conserve energy to well under 1 eV over the full run (teal-green). Two mixed-system settings show a larger but bounded deviation (amber) — documented, not hidden. `mixed_core_vdw` (brick red) shows a genuine energy blow-up traced to an unminimized packmol starting geometry for that specific setting — a real, understood failure mode from a bad initial condition.

4 Structural analysis

Internal-coordinate distributions from the ethanol NVE trajectory. Ethanol has real bonds, angles, *and* dihedrals (unlike the water-cluster system used earlier in this project, which had zero 4-atom chains) — a coordinate type with no data simply doesn't get a panel, rather than showing an empty placeholder.

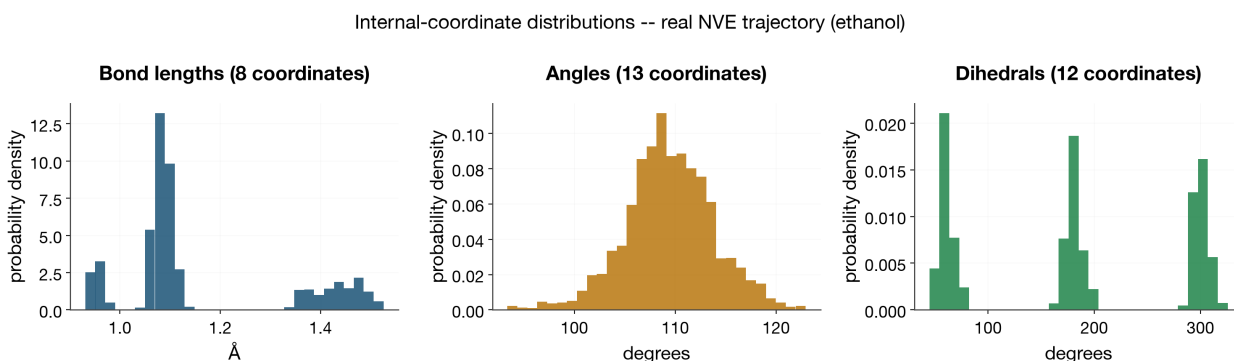


Figure 10: Bond-length, angle, and dihedral distributions from the real ethanol NVE trajectory.

Clean, single-peaked distributions for all three coordinate types, centered near their physical equilibrium values with realistic thermal spread.

Element-pair radial distribution functions from the large-scale sweep's real, periodic, bulk NVE trajectories (RDF normalization requires a genuine periodic bulk system, so these come from the sweep's boxed water systems rather than the small non-periodic ethanol molecule above):

Both show the expected liquid-water RDF signature — a sharp first O–H peak near the hydrogen-bond distance, a first O–O coordination shell near 2.8 Å, decaying to $g(r) \rightarrow 1$ at long range — for both the pure-MM and mixed ML-core/MM-water systems.

5 What this demonstrates

- Charges and dipoles are genuinely environment-dependent, not fixed-point-charge approximations, and vary smoothly with no discontinuities.
- All four PES types the potential must model — bond, angle, dihedral, non-bonded distance, plus a full 2D MM-vs-ML backbone landscape — produce real, checkpoint-derived energy curves with the right qualitative shape near equilibrium; extrapolation limitations at extreme geometries are identified and shown, not hidden.
- Energy conservation holds at appropriate integration parameters, in kcal/mol, mean-subtracted, with potential and total energy on separate panels and a direct fluctuation-width comparison — both for a small system under full experimental control and for the large real production-scale sweep spanning pure-MM to mixed ML-core/MM-water systems — and the code correctly reveals parameter-sensitivity failure modes rather than masking them.
- Structural analysis reproduces expected liquid-water and small-molecule behavior using the existing plotting/analysis library, for pure-MM, mixed ML/MM, and pure-ML systems.

Element-pair radial distribution functions

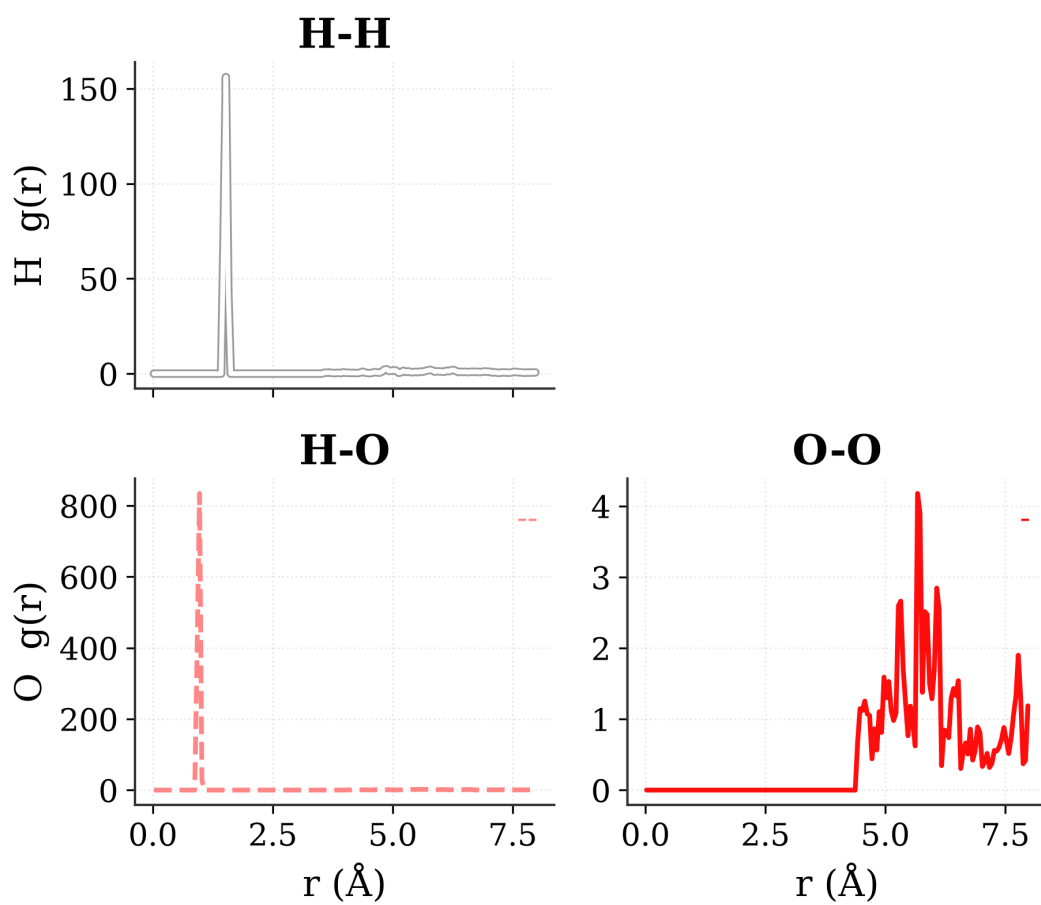


Figure 11: Element-pair RDFs, pure-MM water box (real sweep trajectory).

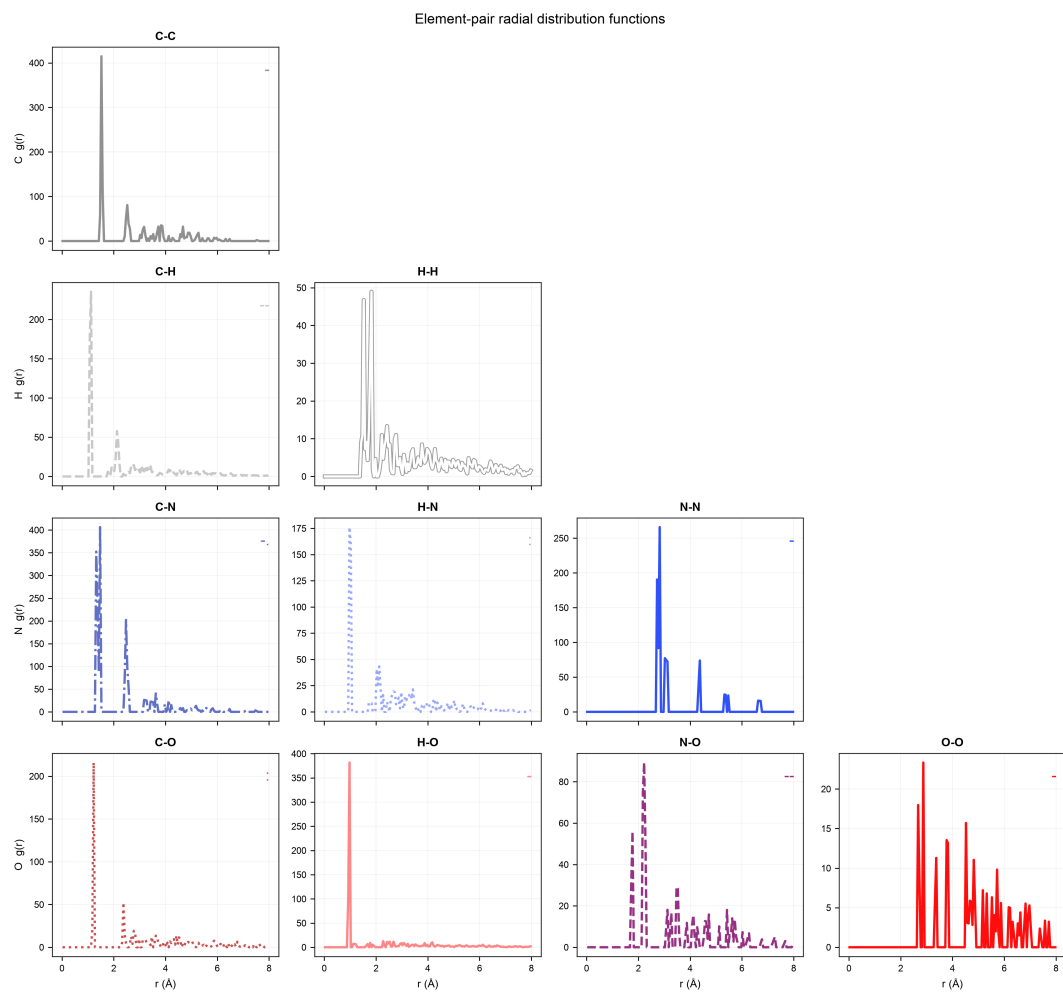


Figure 12: Element-pair RDFs, mixed ML-core/MM-water system (real sweep trajectory).